

Linear systems and elimination

Optimal uniform row-deletion singular-value constant

Difficulty: challenging **Importance:** interesting to the community**Status:** Solved

Independently reviewed resolution - 2026-09-11

Affirmative sharp-constant resolution. Theorem 2 and Sections 6-7 prove the sharp constant $c_\theta = \sqrt{h_\theta}$. The upper bound is uniform over all unit-row matrices and even all $m \geq 1$ for sufficiently large n , with squared normalized error $O_\theta(n^{-1/6})$. Theorem 1 supplies matching spherical realizations along every high-aspect-ratio sequence, proving the exact eventual-uniform optimality quantifiers.

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The difficulty, importance and rating rationale below are historical assessments of the original open target. Original statements, references and dated audits are preserved.

Rating rationale: Challenging reflects an optimal uniform extremum over matrices and deleted row sets; community impact is a sharp robustness limit for row-sampling methods.

Problem statement

Fix $0 < \theta < 1$. For a real $m \times n$ matrix A with every row of Euclidean norm one, put

$$s_\theta(A) = \min_{S \subseteq \{1, \dots, m\}, |S| = \lfloor \theta m \rfloor} \min_{\|x\|_2=1} \|A_S x\|_2, \quad M_{m,n}(\theta) = \sup_{\|a_i\|_2=1} \sqrt{\frac{n}{m}} s_\theta(A).$$

Here a_i is row i of A , and A_S contains exactly the rows indexed by S .

Determine the sharp asymptotic upper constant for $M_{m,n}(\theta)$ as $n \rightarrow \infty$ and $m/n \rightarrow \infty$. In particular, is it

$$c_\theta = \left(\frac{1}{\sqrt{2\pi}} \int_{-a_\theta}^{a_\theta} t^2 e^{-t^2/2} dt \right)^{1/2}, \quad \mathbb{P}(|G| \leq a_\theta) = \theta, \quad G \sim N(0,1)?$$

Precisely, the proposed uniform inequality asks whether, for every $\varepsilon > 0$, there exist integers N and R such that

$$M_{m,n}(\theta) \leq c_\theta + \varepsilon \quad \text{whenever } n \geq N \text{ and } m/n \geq R,$$

and whether no smaller constant has this property. The supremum and quantifiers specify the source's uniform $1 + o(1)$ bound; the floor convention handles row-count integrality.

Connection to numerical linear algebra

This asks how well any unit-row linear system can retain its smallest singular value under worst-case row removal, a geometric constraint on robust iterative solvers.

References

1. S. Steinerberger, *Quantile-based Random Kaczmarz for corrupted linear systems of equations*, Information and Inference **12**(1) (2023), 448–465, §2.3, the third question immediately after equation (8). [Published paper](#). [Author preprint](#), v1, §2.3, paragraph following (◇).
2. E. Battaglia, J.-F. Cai, J. Chen, A. Ma, D. Needell, and T. Wu, *Quantile Randomized Kaczmarz for Streaming Linear Systems with Massart Noise*, arXiv:2608.27968v1 (2026), §1.1 (background on the same restricted-singular-value heuristic and the different streaming model), §5. [Preprint](#).

Earlier status check — 2026-09-08

Checked the published question and latest arXiv records on 2026-09-08. Targeted searches for “Steinerberger subsingular”, “small subsingular values”, “quantile Kaczmarz optimal constants”, “unit rows smallest singular value”, and 2025/2026 variants found no resolution of the all-matrices extremum. Recent streaming Kaczmarz guarantees do not supply this universal extremal inequality. The random-ensemble limit in the same source is a related but distinct question: it specifies a random construction, whereas this question optimizes over all unit-row matrices. Openness remains subject to the limits of this search.

Audit update — 2026-09-10

Rechecked Steinerberger’s [primary formulation](#), §2.3. Searches for the optimal row-deletion constant and quantile-Kaczmarz follow-ups found no proof of this extremal value. The new [Cai–Chen–Ma–Wu paper](#), SIAM J. Matrix Anal. Appl. **47** (2026), 802–823, proves a tight QRK subsample-size result, not the universal matrix constant asked for here.